

Conductivity sensor

Conductivity Sensor 5990 is a compact fully integrated sensor for measuring the electrical conductivity of seawater. Designed to be used as stand-alone sensor.

Conductivity measurements combined with temperature and depth provide accurate salinity estimates.

Salinity - the concentration of dissolved salts - directly influences key seawater properties such as density and sound speed. Aanderaa Conductivity sensors use an inductive principle and represent the fourth generation of this technology, ensuring high accuracy and long-term stability.



Easy integration

Easy integration with most loggers and systems



Output formats

RS-232 / RS-422 / RS-485 (Modbus): Direct output in engineering units – no post-processing required



Design

Rugged design with low maintenance requirements



Conductivity sensor 5990 specifications

Conductivity

Range	0–7.5 S/m (0–75 mS/cm)		
Resolution	0.0002 S/m (0.002 mS/cm)		
Accuracy ¹	5990C: ±0.0004 S/m (±0.004 mS/cm)	5990B: ±0.0018 S/m (±0.018 mS/cm)	5990A: ±0.005 S/m (±0.05 mS/cm)
Response time (63%)	< 3 sec ²		
Typical drift	< 0.0001 S/m per month (0.001 mS/cm per month)		

Temperature

Range	-5 to +40°C (23–104°F) ³		
Resolution	0.001°C (0.0018°F)		
Accuracy	5990C: ±0.003°C (0.0054°F) ⁴	5990A/5990B: ±0.05°C (0.032°F)	
Response time (63%)	< 2.6 sec		

Output format

5990/5990U: RS-232
5990R: RS-422
5990M: RS-485

Sampling interval

1 sec–255 min

Output parameters

Conductivity, Temperature, Salinity, Density and Sound of Speed, Raw data

Current drain

Average	0.16 + 48 mA/S where S is sampling interval in seconds
Maximum	100 mA
Quiescent	0.16 mA

Operating depth

Shallow Water	Intermediate Water	Deep Water
0–300 m (0–984.3 ft)	0–3000 m (0–9843 ft)	0–6000 m (0–19690 ft)

Supply voltage

5 to 14VDC

Electrical connection

8-pin Subconn MCB8M

Weight

280 g (8.466 oz)

Dimensions (W x D x H)

36 x 39 x 122 mm (1.4" x 1.5" x 4.8")

Materials

Epoxy coated titanium

Accessories (not included)

Resistor Set 3719 for functional test
Sensor Cable 5335

¹ Calibrated in 208 points.

² Dependent on flow through cell bore.

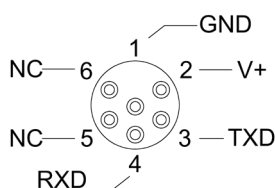
³ Calibrated range is 0 to 36°C (32–96.8°F).

⁴ Only for interval >= 30s

Internal pressure never exceeds atmospheric pressure therefore electronics and sensors are unaffected by deployment depth. The above specifications are for the stand-alone sensor only, not the installation it is utilized with.

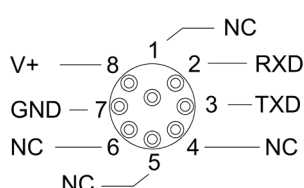
PIN CONFIGURATION 5990U

Subconn MCBH6M-TI



PIN CONFIGURATION 5990

Subconn MCBH8M-TI



PIN CONFIGURATION 5990R/5990M

Subconn MCBH8M-TI

